AICR · OPEN PROBLEM

Exact Full Indifferentiability of 6-Round Feistel Networks

Status. Statement: AI-written, not yet formalized.

Category. Symmetric-Key Cryptography.

The Feistel construction is a standard mechanism for converting public random functions into public pseudorandom permutations. While 5-round Feistel is provably non-indifferentiable from a random permutation (Coron, Patarin & Seurin, CRYPTO 2008), and 8-round Feistel is fully indifferentiable (Dai & Steinberger, CRYPTO 2016), the exact full indifferentiability status of 6-round Feistel remains one of the longest-standing open problems in symmetric-key theoretical cryptography.

1 Formal Setup

Let $\mathcal{F}_n = \{F : \{0,1\}^n \to \{0,1\}^n\}$ denote the set of all n-bit to n-bit functions, and let $\mathcal{P}_{2n} = \mathrm{Perm}(\{0,1\}^{2n})$ denote the set of all $2n$-bit permutations.

Definition 1 (Feistel Network). Given r round functions $\mathbf{F} = (F_1, \ldots, F_r) \in (\mathcal{F}_n)^r$, the r-round Feistel construction $\Psi^r[\mathbf{F}] : \{0,1\}^{2n} \to \{0,1\}^{2n}$ maps $(L_0, R_0) \mapsto (L_r, R_r)$ via the iterative update rule for $i = 1, \ldots, r$:

$$L_i = R_{i-1}, \qquad R_i = L_{i-1} \oplus F_i(R_{i-1}). \tag{1}$$

Definition 2 (Indifferentiability Framework (Maurer, Renner & Holenstein, TCC 2004)). A construction $C^{\mathbf{F}}$ using oracle access to ideal components $\mathbf{F}$ is **indifferentiable** from an ideal primitive $\mathcal{P}$ if there exists a stateful PPT simulator $\mathcal{S}$ with oracle access to $\mathcal{P}$ such that for every PPT distinguisher

$\mathcal{D}$:

$$\mathbf{Adv}^{\text{indiff}}_{C,\mathcal{S}}(\mathcal{D}) := \left|\Pr\left[\mathcal{D}^{C^{\mathbf{F}},\mathbf{F}}(1^n)=1\right] - \Pr\left[\mathcal{D}^{\mathcal{P},\mathcal{S}^{\mathcal{P}}}(1^n)=1\right]\right| \le \epsilon(n), \tag{2}$$

where $\epsilon(n)$ is negligible in the security parameter n.

2 The 6-Round Open Conjecture

Conjecture 1 (Full Indifferentiability of 6-Round Feistel). *Let* $\mathbf{F} = (F_1,\ldots,F_6) \xleftarrow{\$} (\mathcal{F}_n)^6$ *be independent random functions, and let* $\mathcal{P} \xleftarrow{\$} \mathcal{P}_{2n}$ *be a uniform random permutation. There exists an efficient simulator* $\mathcal{S}^{\mathcal{P}}$ *making at most* $\text{poly}(q)$ *queries to* $\mathcal{P}$ *such that for any distinguisher* $\mathcal{D}$ *making at most* q *total queries to its oracles:*

$$\mathbf{Adv}^{\text{indiff}}_{\Psi^6,\mathcal{S}}(\mathcal{D}) \le O\left(\frac{q^k}{2^n}\right), \tag{3}$$

for some universal polynomial degree $k \ge 2$.

3 Current Bounds and Theoretical Barrier

Rounds (r)	Status	Security Bound	Reference
$r \le 4$	Attackable	Non-Indifferentiable	folklore, *a fortiori* from the $r=5$ attack below
$r = 5$	Attackable	Non-Indifferentiable	Coron, Patarin & Seurin (2008)
r = 6	**OPEN**	**Public / Sequential Indiff. Only**	**Mandal, Patarin & Seurin (2012)**
$r \ge 8$	Proven	$O(q^8/2^n)$	Dai & Steinberger (2016)

Table 1: Indifferentiability status of r-round Feistel networks.

Key Open Question: Can a simulator construct consistent responses for queries $F_1,\ldots,F_6$ without incurring exponential blowup

in query history when handling middle-round path completions (F_3, F_4), or does a non-trivial distinguishing attack exist for $r = 6$?